Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Run CNN-LSTM Attention Model
2. Abbreviated Title	Run Model Inference
3. Use Case Id	UC-11
4. Actors	ML Engine (Supporting)
5. Description	The ML Engine runs forward inference through the hybrid CNN-LSTM-Attention architecture using preprocessed inputs (normalized vector + GAF image) to produce a binary heart disease risk probability score.
5.1. Pre Conditions	1. Preprocessed data is available from UC-10. 2. Trained model checkpoint is loaded into memory.
5.2. Task Sequence	1. CNN branch processes the GAF image to extract spatial features. 2. LSTM branch processes the feature sequence for sequential patterns. 3. Attention mechanism computes weighted feature importance. 4. Fused representation passes through the classification head. 5. System outputs sigmoid probability and risk label (HIGH/LOW).
5.3. Post Conditions	1. Binary classification output (probability + label) is produced. 2. Internal activations are stored for XAI computation.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)